Meera Iyer

Postdoctoral Researcher, Computational Biology
meera.iyer@example.ac.in | Bengaluru, India
scholar.google.com/citations?user=AbCdEf123 | orcid.org/0000-0002-1825-0097

Summary

Computational biologist working on single-cell transcriptomics and gene-regulatory network inference, with six peer-reviewed publications.

Research Experience

Postdoctoral Fellow, National Centre for Biological Sciences — February 2021 – Present
Bengaluru

- Develop probabilistic models for inferring regulatory networks from single-cell RNA-seq data.
- Built a Nextflow pipeline that processes 1.2 million cells per run on the institute cluster.
- Co-supervise two doctoral students and one master's student.
- Wrote the computational aims of a successful DBT grant proposal worth INR 1.4 crore.
- Maintain the lab's shared reference atlas and its documentation.
- Organise the fortnightly computational journal club.

Research Intern, European Bioinformatics Institute — May 2019 – August 2019
Hinxton, United Kingdom

- Benchmarked six batch-correction methods on 40 public datasets.
- Contributed a Python package for metric computation that is now used by two other groups.

Teaching Assistant, Indian Institute of Science — August 2016 – April 2019
Bengaluru

- Taught tutorials for the graduate course on statistical genomics for three years.
- Designed weekly R programming assignments and graded 60 students per term.

Education

Indian Institute of Science, Bengaluru — August 2015 – December 2020
Ph.D., Computational Biology

University of Hyderabad, Hyderabad — July 2013 – May 2015
M.Sc., Bioinformatics

Mount Carmel College, Bengaluru — June 2010 – April 2013
B.Sc., Biotechnology

Publications

1. Sparse Bayesian inference of gene-regulatory networks from single-cell data. *Nature Communications*, 2023.
2. A benchmark of batch-correction methods for single-cell transcriptomics. *Genome Biology*, 2022.
3. Cell-type specific enhancer usage during zebrafish fin regeneration. *Development*, 2021.
4. Scalable pseudotime estimation with optimal transport. *Bioinformatics*, 2020.
5. Transcriptional dynamics of the early immune response in Indian major carp. *BMC Genomics*, 2019.
6. An open atlas of regulatory elements in rice. *Nucleic Acids Research*, 2018.

Awards and Fellowships

- DBT Junior Research Fellowship (2015)
- Best Poster Award, ISMB 2019

Computational Skills

Languages: Python, R, Bash
Workflows: Snakemake, Nextflow
Machine learning: PyTorch, scikit-learn
Infrastructure: Linux, SLURM, Docker

Service and Outreach

Mentor, Women in Bioinformatics India — January 2021 – Present

- Mentor undergraduate women applying to computational biology programmes.

Volunteer Explainer, Bengaluru Science Gallery — March 2017 – February 2018

- Explained genomics exhibits to school groups on weekends.

Professional Memberships

- Member, International Society for Computational Biology
- Member, Indian Society of Cell Biology